# Kimia Ziaei

kimia.8102.z@gmail.com kimia.ziaei81@sharif.edu +98 (910) 845-5100

## Education

**Sharif University of Technology** *Oct 2021 – July 2026*
*Bachelor of Science in Industrial Engineering*

- CGPA: 19.49/20 (4/4)
- Ranked 1st out of 85 students

**Sharif University of Technology** *Aug 2023 – July 2026*
*Minor in Economics*

- CGPA: 19.40/20 (4/4)

**Selective Public High School** *Jun 2018 – Jun 2021*
*Diploma in Physics & Mathematics*

- CGPA: 19.90/20 (4/4)

## Experience

**1&1 Company** *Jul 2024 – Sep 2024*
*International Marketing Intern*

- Collaborated with Prof. Majid Rafiee and Dr. Ghazimoradi on agricultural sector projects.
- Supported 1&1 in exploring international export and production opportunities, including identifying potential partners and suppliers, analyzing market trends, and providing regulatory and logistical guidance.
- Gathered and processed agricultural datasets from FAO, EFSA, and various national Ministries of Agriculture for analysis.

## Notable Projects

**Income Inequality and Crime Patterns Across U.S. States** 2024
*Econometrics Course Project – Lecturer: Dr. Naser Amanzadeh*

- Compiled a comprehensive state-level dataset combining crime statistics and income distribution metrics from official sources.
- Applied econometric modeling techniques, including OLS and fixed-effects regressions, to examine relationships between inequality and crime.
- Investigated the influence of income disparity (Gini coefficient) on different types of crime, with rigorous robustness testing.
- Interpreted outcomes using established theories in economic sociology and public policy.

**Simulation of Patient Flow in a Hospital** 2024
*Fund of Comp Simul Course Project – Lecturer: Dr. Nafiseh Sedghi*

- Designed a discrete-event simulation capturing patient movements through triage, labs, operating rooms, ICU/CCU, and general wards.
- Implemented stochastic patient pathways and resource constraints using Python.
- Fine-tuned arrival and service parameters, validating model accuracy through tracing and system stability analysis.
- Monitored performance metrics including wait times, queue lengths, bed occupancy, and overall patient throughput.
- Conducted scenario analyses to evaluate capacity adjustments and improve operational efficiency.

**Optimizing NYC Green Taxi Operations** 2024
*Transportation Planning Course Project – Lecturer: Dr. Erfan Hasannayebi*

- Processed large-scale NYC taxi datasets to generate spatial and temporal demand features.
- Built predictive models using Random Forest and XGBoost to forecast demand at the zone-hour level.

- Formulated and solved an optimization model for vehicle distribution and repositioning across zones using Pyomo and Gurobi.
- Compared optimized fleet management strategies against historical performance metrics.
- Provided actionable insights to enhance operational decision-making and reduce idle time.

**Mixed-Integer Programming for Multi-Machine Production Scheduling** 2024
*Operations Research II Course Project – Lecturer: Dr. Maryam Radman*

- Modeled a production scheduling problem.
- Implemented the MILP model in Python (Pyomo and Gurobipy) and GAMS.
- Optimized product sequencing to minimize total completion time while satisfying machine-specific delays and constraints.

**NBA Ticket Sales Optimization** 2023
*Operations Research I Course Project – Lecturer: Dr. Kourosh Eshghi*

- Developed a mathematical model for ticket allocation and pricing for a 16-team NBA tournament with two ticket types.
- Implemented the model in Python to simulate ticket sales under capacity and operational constraints.
- Performed sensitivity analysis to evaluate the impact of pricing strategies and ticket allocation on total revenue.

## Teaching Assistantships

**Probability Theory & Its Applications** Fall 2024, Spring 2025
*Lecturer: Dr. Nafiseh Sedghi*

- Conducted weekly tutorial sessions and managed quiz preparation, administration, and grading.

**Operations Research II** Spring 2025
*Lecturer: Dr. Maryam Radman*

**Queueing Theory and Probabilistic models** Spring 2025
*Lecturer: Dr. Maryam Radman*

**Macroeconomics** Spring 2025
*Lecturer: Dr. Reza Bakhshiani*

- Conducted tutorial sessions, prepared lecture slides, and designed, administered, and graded assignments, quizzes, midterms, and final exams.

**General Economics** Fall 2024
*Lecturer: Dr. Mohammad Vesal*

- Conducted weekly tutorial sessions and designed, administered, and graded assignments, quizzes, midterms, and final exams.

**Engineering Statistics** Fall 2024
*Lecturer: Dr. Zahra S. Hasanpour Jesri*

**Statistical Quality Control** Fall 2024
*Lecturer: Dr. Majid Rafiee*

**Information Technology** Fall 2023
*Lecturer: Dr. Moslem Habibi*

**Engineering Economy** Fall 2023
*Lecturer: Dr. Maryam Rezapour Niari*

## Voluntary Activities

**GAMEIN Event, Sharif University** *Jun 2023 – Feb 2024*
*Idea Team Member*

- Contributed to designing a simulation game by determining product categories, inventory and logistics rules,

trading policies, and game setup during team workshops.

**Mentor & Academic Advisor – High School Students** *Jun 2021 – Jun 2023*

- Guided students in study planning, held weekly check-ins, and provided academic support to prepare for university entrance exams.

## Selected Courses

| | | | | | |
|---|---|---|---|---|---|
| Probability & Appl | 19.3/20 | Fun of Comp Simul | 20/20 | General Economics | 19.3/20 |
| Operation Research I | 19.7/20 | Prob Models & Queueing | 20/20 | Microeconomics | 20/20 |
| Operation Research II | 19/20 | Transportation Planning | 20/20 | Macroeconomics | 20/20 |
| Decision Making | 19.7/20 | Supply Chain Mngmt Fund | 20/20 | Econometrics | 20/20 |
| Eng Statistics | 19.5/20 | Linear Algebra | 20/20 | Fund of Game Theory | 17.7/20 |
| Project Control | 20/20 | Convex Optimization | 19.6/20 | Financial Economics | 19.2/20 |

## Skills

**Programming:** Python, GAMS, Stata
**Miscellaneous:** Power BI, Visual Paradigm, AnyLogistix, Microsoft Office, LaTeX

## Languages

**Persian**: Native

**English**: Fluent (TOEFL iBT: 109; R30, L30, S22, W27)

## References


**Kourosh Eshghi**
Professor
Department of Industrial Engineering
Sharif University of Technology
Email: eshghi@sharif.edu

**Nafiseh Sedghi**
Assistant Professor
Department of Industrial Engineering
Sharif University of Technology
Email: nafiseh.sedghi@sharif.edu

**Mohammad Vesal**
Associate Professor
Graduate School of Management and Economics
Sharif University of Technology
Email: m.vesal@sharif.edu

**Maryam Radman**
Assistant Professor
Department of Industrial Engineering
Sharif University of Technology
Email: radman@sharif.edu